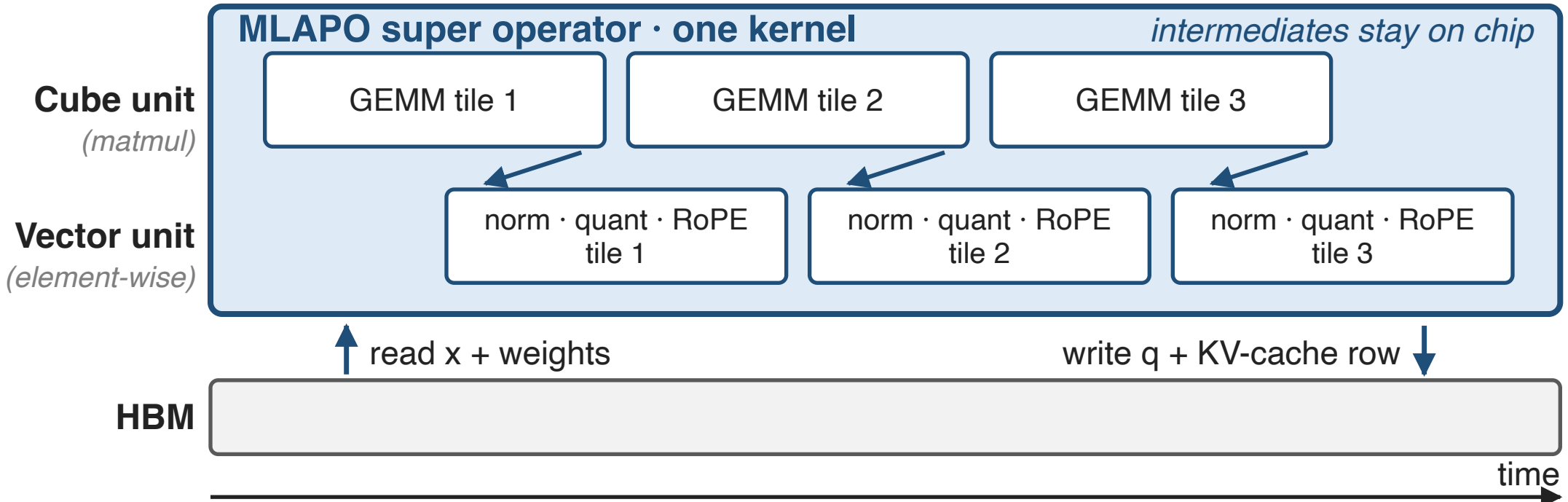
MLAPO super operator · one kernel
intermediates stay on chip
Cube unit
(matmul)
GEMM tile 1
GEMM tile 2
GEMM tile 3
Vector unit
(element-wise)
norm · quant · RoPE
tile 1
norm · quant · RoPE
tile 2
norm · quant · RoPE
tile 3
read x + weights
write q + KV-cache row
HBM
time